

A replicated quasi-experimental study on the influence of personality and team climate in software development

Marta N. Gómez · Silvia T. Acuña

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Abstract This article reports a replication of a quasi-experimental study analyzing how personality factors and team climate influence software development team effectiveness, product quality and team member satisfaction. The replication was designed on the basis of the original quasi-experimental study, both of which were run in an academic setting. In the original study, data were collected from a sample of 35 three-member developer teams. All these teams used an adaptation of extreme programming (XP) to the academic environment to develop the same software system. In the replication, the data were collected from a sample of 34 three- or four-member developer teams working on the same software project. Student teams used a common object-oriented software development paradigm to solve the set problem and applied the Unified Process. In both studies all teams were formed at random, and their members were blind to the quasi-experimental conditions and hypotheses. The replication of this empirical study aims to verify the results of the original quasi-experiment. It examines, first, whether personality factors (neuroticism, extroversion, agreeableness, openness to experience and conscientiousness) are related to the quality of the developed software and team member satisfaction and, second, whether the preferences, perceptions and preferences-perceptions fit for the four team climate factors (participative safety, support for innovation, team vision and task orientation) are related to the quality of the developed software and team member satisfaction. The results of the replication corroborate some of the findings of the original study. On the one hand, the results revealed that there is a significant correlation between the extroversion personality factor and software

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M. N. Gómez

Escuela Politécnica Superior, Universidad CEU San Pablo, Campus de Montepríncipe 28668 Boadilla del Monte, Madrid, Spain
e-mail: mgomez.eps@ceu.es

S. T. Acuña (✉)

Escuela Politécnica Superior, Universidad Autónoma de Madrid, Calle Francisco Tomás y Valiente 11, 28049 Madrid, Spain
e-mail: silvia.acunna@uam.es

quality, but no significant correlation between the extroversion personality factor and team satisfaction. Also, we found that the perception of team climate where participative safety is high is related to better quality software. We observed significant relationships between the perception of the four team climate factors and team member satisfaction. Additionally, the results showed a positive relationship between software quality and teams in which the real climate perception at the end of the project is better than preferences stated by team members at the outset of the project for the participative safety factor. Finally, we found that teams where the real climate is better than the stated preferences for the team orientation factor exhibit a direct and positive relationship to team member satisfaction.

Keywords Personality factors · Team climate · Software quality · Team satisfaction · Team building · Quasi-experimental replication

1 Introduction

The software engineering community recognizes that software process effectiveness largely depends on human and social factors (Boehm et al. 2000). Some investigations on people in software development (Peslak 2006; Sfetsos et al. 2009; Yang and Tang 2004) assume that personality, cohesion and conflict are factors that influence team performance. Cruz et al. (2011) report a systematic review of four decades of research on the influence of personality in software engineering. They considered that the study of personality can help people improve job performance. They found that over 80 % of the analyzed papers were empirical, and, of these, almost 40 % used experiments or quasi-experiments as a research method. In their paper, they suggested that more replication should be conducted to corroborate and consolidate the findings and improve the transfer of results to the software industry. As many are standalone experiments, it is not possible to extract robust conclusions and make generalizations. Nowadays, experimental software engineering (ESE) research has evolved from individual experiments towards families of experiments. Researchers have moved on from independently running laboratory experiments to collaborative and global research, defining coordinated experiments (Selby 2007) and linking up sets of experiments (Pfleeger 2005).

The software development process is composed of interrelated activities, demanding that stakeholders liaise and communicate adequately to successfully develop the project. On this ground, software engineers are required not only to have technical knowledge but also to be able to work as a team to develop top quality software. This calls for a team climate that is propitious to the performance of these activities (Dybå and Dingsøyr 2008; Macias et al. 2003). These papers compare development processes (agile vs. traditional methods) and examine factors such as developer team cohesion, job satisfaction and team productivity. Goparaju et al. (2011) classified the factors affecting software development team performance and stressed the soft (non-technical) factors affecting the performance of these teams. They found that soft factors, such as team climate, team diversity, team innovation, team member competencies and characteristics, top management support and team leader behavior, have an effect on software development team performance. Mutual trust and effective communication are prioritized factors affecting software development team performance. Accordingly, knowledge of software development team climate preferences and perceptions, cohesion, conflict and task characteristics can be important tools for improving software engineering practice.

Social psychology, on the other hand, has analyzed some of the factors that can affect team performance, including team member personality factors and their relationship to task

characteristics (Hackman and Oldham 1980; Barrick and Mount 1991; Barry and Stewart 1997; Molleman et al. 2004). But team performance also depends on the interactive effects of team climate characteristics (Burch and Anderson 2004) and other team behavior factors, like interactions between people (conflict and cohesion). Team climate affects personal relationships within teams. Such relationships are key factors for the success or failure of these teams and the work they do (Zander 1994; Jehn 1997; Mañas et al. 1999; Curral et al. 2001). Although some trends appear to apply to all branches of activity equally, the results of these studies on team formation depend on the activities that they perform. Therefore, we think that software development teams should be studied separately. This will be useful for determining which factors influence the set of software development activities in particular.

The original quasi-experiment was designed to determine some of the factors that are related to software quality and software development team satisfaction. This study was conducted in an academic setting (Acuña et al. 2008, 2009) and primarily analyzed personality factors and team climate factors in order to investigate their influence on software quality and perceived team member satisfaction. The general goals of this original quasi-experiment were to determine the relationships of:

- Team member personality factors (neuroticism, extroversion, agreeableness, openness to experience and conscientiousness)
- Team climate during the software construction process (participative safety, support for innovation, team vision and task orientation preferences, perceptions and preference/ perception fit)
- Team processes during the software construction process (task conflict, social conflict, cohesion)
- Characteristics of the task performed by the team (interdependency and autonomy)

with respect to developed software quality and team member satisfaction.

The findings of the original study with respect to these relationships were not conclusive. This motivated several replications of the original study in order to corroborate the results and arrive at more robust findings.

The aim of this paper is to describe the replication of empirical research applying the guidelines defined for reporting experimental replications proposed by Carver (2010). Replications, reports describing replications by other researchers, as well as the aggregation of replication results will be useful for software engineers to draw conclusions and consolidate findings that would otherwise be out of the question.

This paper reports the replication of the original quasi-experiment developed to ascertain some of the software development team factors that are related to the quality of the developed software and team satisfaction. It is important to determine these factors in order to acquire recommendations to improve team formation, the quality of the developed software product, team climate and software engineer satisfaction. The design of the empirical study was divided into two parts:

- The first part was concerned with analyzing the relationships of developer team personality factors, team processes and task characteristics with respect to the quality of the developed software product and team member satisfaction.
- The second part focused on the study of the team climate factor preferences and perceptions (Anderson and West 1998, 1999; Anderson and Burch 2003). This part also examined whether the fit between software developer team climate preferences and perceptions is related to product quality and software development team member satisfaction in this experimental study.

The remainder of the paper is organized as follows. Section 2 provides information on the original study that is useful for understanding the replication. Section 3 describes the replication. Section 4 compares the results of the replication and the original study. Section 5 discusses the conclusions of both studies and the resulting recommendations on team formation. Finally, Section 6 outlines the final conclusions of the article after having applied the experimental replication reporting guidelines.

2 Information About the Original Study

The original study was about teams and the collective effort of individuals working together towards the common goal to build a software product. The original study set to analyze some social psychology factors that have been seen to influence team effectiveness and extend the results by analyzing the relationships between such factors and the impact on software development teams applied to software engineering.

The empirical study run fits the description of a quasi-experiment (Cook and Campbell 1979). Quasi-experiments are run when the subjects cannot be randomly assigned to an experimental condition or, alternatively, a treatment cannot be randomly assigned to a group. In other words, the group applies a given treatment, but this treatment is not allocated at random. A quasi-experiment is characterized by being neither very intrusive nor very costly. Statistical tests, like correlations, regressions, mean differences analysis, variance analysis, etc., are applied to the collected data, as applicable. In quasi-experiments, however, it is hard to satisfy the conditions required to establish a causal relationship between independent and dependent variables. This is a correlational study.

As mentioned in the Section 1, the study was divided into two parts. Part one addressed personality factors, team processes and task characteristics, and part two examined team climate factor preferences and perceptions. Both parts of the original study have been published, not so the replication or the comparison of the two experimental studies. The first part of the original study concerning the relationships between team personality factors and software product quality and team member satisfaction is reported in Acuña et al. (2009). The original study measures team member personalities based on the Big Five personality factors (Costa and McCrae 2002):

- Neuroticism is a broad dimension that includes traits like anxiety, moodiness, irritability or frustration.
- Extroversion is a trait of people who take a trusting and enthusiastic view of others, which is associated with being sociable, assertive and talkative.
- Agreeableness is a trait of showing altruistic concern and emotional support towards other people.
- Openness to experience denotes inquisitiveness about new ideas, values, feelings and interests.
- Conscientiousness is a trait of perseverant, scrupulous and responsible behavior.

The original study covered another important component worth taking into account, namely, the characteristics of the task that the teams perform. According to research by Molleman et al. (2004), the key task characteristics influencing team behavior are:

- Interdependency, which refers to the situation in which the process and the outcome of a task affect the process and outcome of other tasks. This means that the decisions on goals, work methods and delivery schedules taken by a team member affect his or her colleagues' decisions and vice versa;

- Autonomy, which refers to how free a team is to make decisions on goals (what), work methods (how), delivery schedules (when) and work distribution among team members (who).

Finally, there is another important component for gaining a better understanding of these relationships. This is group behavior (Barrick et al. 1998; Yang and Tang 2004):

- Task conflict, which occurs when people have different opinions about which is the best process or option for performing an activity. This type of conflict can be a positive force for the team, pushing team members to look for different ways of doing things. The resolution of task conflicts leads to innovation and makes change more acceptable.
- Social conflict, which refers to incompatible feelings and emotions among team members. This type of conflict leads to personal clashes that are an obstacle to satisfactory activity performance and have a negative impact on team work.
- Cohesion, which is a measure of the group unity. It is the result of the inclination of team members to stay with the group. Team cohesion is an indicator of its strength. Additionally, cohesion is a sign of team member commitment to the team.

Acuña et al. (2008) reports the second part of the quasi-experiment focusing on the study of the relationships between team climate factors and software product quality and team member satisfaction. Team climate is a complex concept and has to be decomposed into dimensions. On the one hand, there are each team member's team climate preferences, that is, the climate in which they would like to work. The Team Selection Inventory (TSI) psychometric test is used to measure preferences (Anderson and Burch 2003; Burch and Anderson 2004). On the other hand, there are the team climate perceptions, that is, the real team climate. The Team Climate Inventory (TCI) psychometric test is used to measure perceptions (Anderson and West 1994, 1998, 1999). Both questionnaires are closely related and validated, that is, they are directly comparable. They are applicable to any task and can, therefore, be applied to software development. If used together, these two tests provide an indication of the person-team fit (Anderson and West 1998). The TSI and TCI questionnaires were designed to measure climate considering four key factors for effective team operation and innovation:

- Participative safety: how much trust participating team members feel there is within the group when explaining their opinions and ideas.
- Support for innovation: support provided by the team for innovative ideas.
- Team vision: how clearly the team defines goals.
- Task orientation: how much effort the team puts into achieving excellence in what it does.

The original study set out to examine whether all these independent variables (personality factors, task characteristics, team process and team climate factors) are related to the dependent variables, which are:

- Team member satisfaction, which indicates the extent to which a team member concurs and agrees with his or her teammates with regard to the work method, generated climate, goal achievement, etc.
- Software quality, which refers to the appraisal of a set of aspects considered relevant for establishing the quality level and excellence of the developed software product.

We now describe the goals, hypotheses, variables, subjects, measurement instruments and the quasi-experimental procedure. This report conforms to the guidelines for empirical research in software engineering and experimental replications (Carver 2010) as closely as possible.

2.1 Goals, Hypotheses and Variables

The original study was run on university students that had to develop a software system as part of a course practical assignment. Therefore, the context of the study is a laboratory environment with non-professional participants.

The objectives of the research were to analyze and compare the relationships of:

1. The Big Five personality factors (neuroticism, extroversion, openness to experience, agreeableness and conscientiousness) at team level, task characteristics (autonomy and interdependency) and team processes (cohesion, task conflict and social conflict), and
2. Climate preferences, perceptions and preferences-perceptions fit (participative safety, support for innovation, team vision and task orientation) within a team.

with respect to the quality of the developed software and team member satisfaction.

A series of alternative hypotheses were stated for each of these research goals. We used a subindex-based nomenclature to correctly identify the stated hypotheses. The nomenclature format is H_{ab_nn} , defined as follows:

- a: Identifies the goal of the study to which the hypotheses are related. Values 1 and 2 refer to the hypotheses related to the first part (first goal) and second part (second goal) of the study, respectively.
- b: Further details the goals of the study with respect to the response variables. Values 1 and 2 refer to the hypotheses related to the developed software quality and software development team satisfaction, respectively.
- nn: Correlatively numbers the alternative hypotheses of each part of the study and dependent variable. The possible values are 1 to 12.

The alternative hypotheses derived from the first goal with respect to the developed software quality were:

- H_{11_01} : There are relationships between all team member personality factors and developed software quality.
- H_{11_02} : There are relationships between task characteristics, autonomy and interdependency, and developed software quality.
- H_{11_03} : There are relationships between team processes, task conflict, social conflict and cohesion, and developed software quality.

The alternative hypotheses derived from the first goal with respect to software development team satisfaction were:

- H_{12_01} : There are relationships between all team member personality factors and software development team satisfaction.
- H_{12_02} : There are relationships between task characteristics, autonomy and interdependency, and software development team satisfaction.
- H_{12_03} : There are relationships between team processes, task conflict, social conflict and cohesion, and software development team satisfaction.

The alternative hypotheses derived from the second goal with respect to developed software quality were:

- H_{21_01} to H_{21_04} : The team climate preference for [participative safety | support for innovation | team vision | task orientation] is related to a better quality software product.

- H_{21_05} to H_{21_08}: The team climate perception for [participative safety | support for innovation | team vision | task orientation] is related to a better quality software product.
- H_{21_09} to H_{21_12}: The [participative safety | support for innovation | team vision | task orientation] preferences-perceptions fit is related to the quality of the software produced, such that, at the end of the project, the teams with better than or equal to preferred [participative safety | support for innovation | team vision | task orientation] produce a better quality software product than teams with worse than preferred climate.

The alternative hypotheses derived from the second goal with respect to software development team satisfaction were:

- H_{22_01} to H_{22_04}: The team climate preference for [participative safety | support for innovation | team vision | task orientation] is related to software development team satisfaction.
- H_{22_05} to H_{22_08}: The team climate perception for [participative safety | support for innovation | team vision | task orientation] is related to software development team satisfaction.
- H_{22_09} to H_{22_12}: The preferences-perceptions fit [participative safety | support for innovation | team vision | task orientation] is related to software development team satisfaction, such that, at the end of the project, the teams with better than or equal to preferred [participative safety | support for innovation | team vision | task orientation] achieve better satisfaction than teams with worse than preferred climate.

Table 1 summarizes the independent and dependent variables analyzed in the quasi-experiment.

The values of the independent and dependent variables were all taken directly from questionnaires (see Section 2.3), except the independent variable climate preferences-perceptions fit and the dependent variable software product quality. The climate preferences-perceptions fit was measured by calculating, for each of the four team climate factors, the mean differences between the mean preferences before starting team work and the mean perceptions at the end of development. These differences were classed according to three fits: better than preferred climate (team members perceived the real climate to be better than preferences), equal to preferred climate (members perceived the climate equal to preferences) and worse than preferred climate (members perceived the real climate to be worse than preferences). For each factor, we calculated the 34th and 67th percentiles to separate the data into three parts containing an identical number of teams, i.e., the 34 % that had the lowest difference values (better than preferred); the 33 % that had medium values (equal to preferred) and the 33 % that had the highest values (worse than preferred).

The chosen dependent variable was software product quality. In order to measure software quality, we selected a set of standard software engineering criteria, primarily taken from SWEBOK 2004: decomposition and modularization, testability (the effort it takes to validate the modified software), functionality (software's capability to do what it should, i.e., characteristics relating to achievement of the basic purpose for which the software is being engineered), reusability (the likelihood of a module being able to be used again to add new functionalities with slight or no modification), programming style and team member participation. These criteria (see Table 2) were evaluated on a 1-to-4 point scale, as shown in Table 3. The score for each deliverable was calculated by applying a weighted formula

Table 1 Quasi-experiment independent and dependent variables

Independent variables		Dependent variables
Global	Specific	
Personality	- Neuroticism - Extroversion - Openness to experience - Agreeableness - Conscientiousness	Satisfaction software quality
Task characteristics	- Interdependency - Autonomy	
Team processes	- Task conflict - Social conflict - Cohesion	
Team climate	- Preferences - Perceptions - Preferences-perceptions fit	
	- Participative Safety - Support for Innovation - Team Vision - Task Orientation - Better than preferred - Equal to preferred - Worse than preferred	

(Acuña et al. 2008, 2009) that relates these metrics. Each metric is worth 20 %, and the formula calculates the weighted score on a 0-to-10 point scale.

A common work plan was established for all teams at the start of the course. This work plan included specific milestones indicating when deliverables were to be submitted. These deliverables were evaluated by the professors, and this evaluation served as feedback for students. The final project grade was calculated using a weighted formula, in which the first two deliverables were worth 20 % and the last two were worth 30 %.

This quasi-experiment was performed at team level, and the questionnaires were administered to individuals. On this ground, the data had to be aggregated in order to obtain the team construct. With the exception of the software product quality, which was calculated at team level, team-level variables were generated by aggregating the scores of the members of each team. We used the arithmetic mean as the group aggregation index.

Table 2 Software product evaluation criteria for original study

Criteria (Beck 1999; Bentley 2000; IEEE 2004; Pfleeger 2001)	Metric	Evaluated items
- Decomposition and modularization	- Number of modules and coupling	Code and drafted documents
- Testability	- Number of defects detected by the automatically executed test case set	
- Functionality	- Number of satisfied requirements	
- Reusability	- Number of reused modules	
- Programming style	- Guidelines defined on subject website	Team members
- Participation	- Checklist of laboratory session observations	

Table 3 Scoring of software product quality criteria

Score	Category	Range
1	Poor	Software product satisfied up to 25 % of the aspects for each metric
2	Acceptable	Software product satisfied from 26 % to 50 % of the aspects for each metric
3	Good	Software product satisfied from 51 % to 75 % of the aspects for each metric
4	Excellent	Software product satisfied from 76 % to 100 % of the aspects for each metric

We identified three context variables that we did not intend to investigate but were likely to affect the quasi-experimental results (see Table 4):

- Software development member knowledge and experience of software development. This variable was blocked by randomly assigning participants to teams which are the quasi-experimental subjects. Randomization assured that two or more teams are likely to be equivalent.
- Software development project (functionalities to be implemented by subjects during the experiment). This variable was blocked by setting the same software development project for all teams.
- Knowledge of the proposed software development method. To control this variable and try to establish a similar level of knowledge, all study participants were given general-purpose training on the agile XP method.

2.2 Participants and Subjects

The participants were second-year computing students at the Universidad Autónoma de Madrid's Higher Technical School of Computing during the 2004/05 academic year. These students were taking the Data Structures and Algorithms course. The project set was to design and implement a distributed library management system. Through this project, students were expected to demonstrate their knowledge of how to use and implement a range of abstract data types using different data structures, as well as an ability to analyze and develop software solutions for the stated problem.

A total of 105 students participated in this empirical study, of which 83 were male (79 %) and 22 were female (21 %). Of the students, 75 % (79 students) were aged under 21 years and 25 % (26 students) were from 21 to 30 years of age.

The students were divided into 35 three-member teams who worked together to develop the software. Each three-member development team was a subject of this quasi-experiment. These teams were formed at random and their members were blind to the quasi-experimental conditions and hypotheses. Teams used a common agile methodology

Table 4 Context variables in original study

Controlled variable	Control mode
Previous knowledge and experience of software development	Randomization forming teams by drawing lots
Software development project	Setting same or equally complex project for all teams
Knowledge of proposed software development method	Standardization of knowledge through training

(an adaptation of XP (Beck et al. 2001) to the academic environment) and the C programming language.

2.3 Instrumentation and Data Collection Procedure

The quasi-experimental procedure was divided into three phases:

- The PRE or preliminary phase is the period during which the project development teams are formed but have not yet performed any tasks (the weeks from the start of the course until the start of teamwork). Five personality factors and team climate preferences were measured during this phase.
- The DURING phase is the period during which the teams work on the software project until the project is around 45 % complete. During this period the questionnaires on possible team processes (task conflict, social conflict and cohesion) and team climate perceptions and real work climate are handed out and collected in.
- The POST phase is the period after development when project development is nearing completion (90 % finished). This period coincides with the end of the course after submission of the project developed during the course. The measurements taken in this phase through questionnaires are team climate perceptions, task characteristics (interdependency and autonomy) and level of satisfaction of team members after teamwork. At the end of this stage, we calculate the climate preferences-perceptions fit. The quality of the software built by each team is measured after submission of the software report and product.

The questionnaires and psychological tests that had to be completed in the quasi-experiment are discussed below:

1. NEO-FFI Test (Costa and McCrae 2002) (Spanish version) to measure and assess team member personality traits. This test is composed of 60 questions measuring five personality factors through 12 items for each factor.
2. Gross Cohesion Questionnaire (Stokes 1983) to measure team member cohesion. This test is composed of seven questions.
3. Intragroup Conflict Scale (ICS) (Jehn 1995) to determine task conflict and social conflict. This revised version contains three items for task conflict and three items for social conflict.
4. Van der Vegt et al.'s questionnaire (2001) to measure task interdependency. This questionnaire uses seven items to assess how dependent team members are on their teammates to do their job, how closely they work together, etc.
5. Molleman's questionnaire (2000) to measure task autonomy. This questionnaire is composed of five items.
6. Gladstein's questionnaire (1984) to measure team satisfaction. It is assessed using three items to indicate how satisfied people are with their teammates, teamwork, etc.
7. Team Selection Inventory (TSI) (Anderson and Burch 2003; Burch and Anderson 2004) to measure the preferred team climates. It is composed of 42 items that are used to assess four team climate factors –participative safety, support for innovation, team vision and team orientation–, plus social desirability questions.
8. Team Climate Inventory (TCI) (Anderson and West 1994, 1998, 1999) to measure the real team climates. It is composed of 42 items that are used to assess four team climate factors –participative safety, support for innovation, team vision and team orientation–, plus social desirability questions.

Therefore, the questionnaires to be completed by the participants were distributed across the different quasi-experimental phases as shown in Table 5.

The collected data were analyzed. The statistical techniques applied follow:

1. Analysis of the correlations of the personality factors, task characteristics (autonomy and interdependency), team processes (cohesion, task conflict and social conflict) and team climate factors with respect to team satisfaction and software product quality. The linear correlation or Pearson's coefficient is useful for measuring the strength of the possible relationship between the study variables in each phase of the quasi-experiment. This coefficient is applicable when the possible relationship between the variables is linear. Note that the existence of a correlation among variables does not imply causality, that is, a cause-effect relationship among the two variables.
2. Multiple linear regression among personality factors, social conflict, task conflict, cohesion and task interdependency and autonomy with respect to team satisfaction and software product quality.
3. Linear regressions between each of the team climate factors (climate preferences and perceptions) with respect to team satisfaction and software product quality.
4. Analysis of mean differences among the team climate preferences and perceptions. These differences can be used to categorize the group as a team where members found a better than preferred climate, an equal to preferred climate, or a worse than preferred climate. The analysis of mean differences is also useful for checking the relationship of the team climate fits for each of the four climate factors with respect to software product quality and team satisfaction.

We also used an analysis of variance for split plots within a randomized block design in order to check the significant differences between each of the measurements taken before, during and after project development as a result of teamwork. The detailed analysis was published in Acuña et al. (2008).

Table 6 shows which statistical techniques were used to test the working hypotheses stated for the personality factors, task characteristics and team processes with respect to team member satisfaction and software product quality (goal 1).

Table 7 shows the statistical techniques used to test the working hypotheses stated for goal 2.

Table 5 Original study phases, variables and measurement instruments

Variables	Measurement instruments		
	Pre phase	During phase	Post phase
Personality	NEO FFI test		
Team climate preferences	Team Selection Inventory (TSI)		
Team climate perceptions		Team Climate Inventory (TCI)	Team Climate Inventory (TCI)
Interdependency			Van der Vegt et al.'s questionnaire
Autonomy			Molleman's questionnaire
Cohesion		Gross cohesion questionnaire	
Conflict		Intragroup conflict scale	
Satisfaction			Gladstein's questionnaire

Table 6 Correspondence between statistical techniques and hypotheses for goal 1

Factors	Dependent variable		Statistical technique
	Quality	Satisfaction	
Personality	H _{11_01}	H _{12_01}	Correlations and multiple linear regression
Task	H _{11_02}	H _{12_02}	
Team processes	H _{11_03}	H _{12_03}	

Table 8 lists the roles defined in the original study and the number of people allocated to each role.

2.4 Summary of the Results

This section reports the main results of the analyses and their interpretation for the two goals of the study. The first goal evaluated personality, team processes and task characteristics with respect to software product quality and team member satisfaction. The second goal examined the relationships of team climate with respect to software product quality and team member satisfaction.

The results of the correlation analysis for goal 1 showed that team members are more satisfied when interpersonal relationships (agreeableness) and thoroughness of software development (conscientiousness) are greater. Team satisfaction also increases with team member cohesion. Team member satisfaction drops as both social and task conflict within the team increases. This is understandable taking into account that relationships are more friendly and straightforward in teams where there is no friction among team members. Team participativeness improves the bonds among its members, fortifying team vision and cohesion. People are also likely to find it easier to suggest ideas about how to do their job when the team climate is receptive and innovative.

Table 7 Correspondence between statistical techniques and hypotheses for goal 2

Factors		Dependent variable		Statistical technique
		Quality	Satisfaction	
Climate preference	Participative safety	H _{21_01}	H _{22_01}	Correlations and linear regressions
	Support for innovation	H _{21_02}	H _{22_02}	
	Team vision	H _{21_03}	H _{22_03}	
	Task orientation	H _{21_04}	H _{22_04}	
Climate perception	Participative safety	H _{21_05}	H _{22_05}	Correlations and linear regressions
	Support for innovation	H _{21_06}	H _{22_06}	
	Team vision	H _{21_07}	H _{22_07}	
	Task orientation	H _{21_08}	H _{22_08}	
Climate preference-perception fit	Participative safety	H _{21_09}	H _{22_09}	Mean difference analysis
	Support for innovation	H _{21_10}	H _{22_10}	
	Team vision	H _{21_11}	H _{22_11}	
	Task orientation	H _{21_12}	H _{22_12}	

Table 8 Roles in original study

Role	Description	No. participants
Designer	Decides how treatments are allocated to groups, outlines the operation of the experiment and plans the execution	2 people
Trainer	Transmits treatments to the subjects during training sessions; treatments can be transmitted as part of a course, seminar or workshop	2 people
Monitor	Gives instructions to the subjects about how the session is to be run, hands out materials to subjects, collects in the materials and answer questions	2 people
Measurer	Enacts the measurement procedure on the data gathered during the experiment	1 person
Analyst	Identifies possible outliers, prepares descriptive analyses, and does significance testing, post-hoc comparison testing and other analyses	1 person

Interdependency is positively related to team satisfaction. These are small teams that perform short-term activities in which they are expected to rapidly achieve positive results (very short time to market). Precisely because of this, interdependency is considered vital. If there is no interdependency, frustration about not achieving the set goal would run very high. Agile software development demands such interdependency, where all tasks are closely interrelated and should be completed in very short time periods. Therefore, software team members are mutually dependent.

If there is a conflict among team members, it will logically be harder to reach agreement on how to do the task that they have been set. Autonomy, that is, the option of deciding on how to organize the team and perform tasks, was also found to have a positive relationship with respect to team satisfaction.

Mean team extroversion is the only personality factor that is clearly related to the improvement of the quality of the software developed by teams following an agile development method. Agile software development calls for high interaction among team members. All agile software development team members can be catalogued as project managers and share responsibility for both the success and failure of the developed product. Accordingly, traits such as extroversion, sociability and participation are advantageous for high quality software development.

Table 9 shows the correlations among the personality factors, task characteristics, team processes, team member satisfaction and software product quality that were significant in the quasi-experiment. The correlation coefficients that were significant at a 99 % confidence level are marked with ‘***’ in Table 9. These relationships have a significance level of

Table 9 Matrix of significant correlations between personality, team processes, task characteristics, software quality and team satisfaction in the UAM 0405 quasi-experiment

UAM 0405	Satisfaction	Software quality
Neuroticism		
Extraversion		0.455*
Openness		
Agreeableness	0.334*	
Conscientiousness	0.341*	
Social conflict	−0.354*	
Task conflict	−0.526**	
Cohesion	0.394*	
Autonomy	0.471**	
Interdependency	0.797**	

* $p < 0.5$ ** $p < 0.01$

$p < 0.01$, indicating that the error probability is less than 1 %. Correlation coefficients that identify relationships between variables with a confidence level of 95 % are marked with ‘*’ in Table 9. This indicates that they have a significance level of $p < 0.05$, which means that the error probability is less than 5 %.

Multiple linear regression was another analysis used for study goal 1. It aims to identify the explanatory variables, that is, select the variables that may influence the dependent variable and rule out variables that supply no information. Note that regressions indicate relationships but not their causality.

The multiple linear regression model built for software product quality did not return any variable with a large enough predictive weight. Table 10 shows the result of the multiple linear regression model for satisfaction. We applied a two-step procedure, arriving at a corrected coefficient of determination or correlation R^2 of 0.733 (out of 1). In other words, the model predicts or explains that degree of variance of the team satisfaction variable from task interdependency, cohesion and task conflict. Interdependency ($p = 0.000$) is the variable with the highest predictive power for satisfaction.

The results of the linear regression for goal 2 showed that most preferences for and perceptions of team climate factors before and after the project are not good at predicting software product quality. Only the preference for team vision ($R^2 = 0.134$, $p = 0.030$) before the start of the development project and the perception of participative safety ($R^2 = 0.115$, $p = 0.046$) at the end of the project can explain software product quality. These relationships confirm that teams prefer a climate that promotes teamwork and where the team climate is perceived to be safe, that is, to promote team member participation.

From these results, we can say that there is a relationship between team climate perceptions after software project development and team satisfaction but not between team climate preferences before software project development and team satisfaction. Therefore, teams whose members perceive a climate where all team members are encouraged and moved to participate, which is conducive to innovation and supportive of team work where team members are set clear goals, will be more satisfied with teamwork and will want to stay with the team.

Table 11 illustrates the relationships of software quality and team member satisfaction with respect to the perceptions of the four team climate factors. Table 11 shows the matrix with statistically significant linear regressions (R^2). The R^2 coefficient can be interpreted as the proportion of variance of the software quality and team satisfaction dependent variables explained by the predictors or independent variables (perception of each team climate factor). This is a standard measure whose values range from 0 to 1.

We then conducted an analysis of the mean differences between team climate factor preferences and perceptions (Acuña et al. 2008). These differences were used to establish categories of teams depending on the real climate perceived within the team and the climate

Table 10 Regression model for team satisfaction

	Non standardized regression coefficients		Statistical T	p
	B	Standard error		
(Constants)	8.013		2.475	
Interdependency	0.521	0.067	7.754	0.000
Task conflict	-0.674	0.174	-3.884	0.001
Cohesion	-0.168	0.095	-1.760	0.088

$R^2 = 0.733$

Table 11 Matrix of linear regressions of team climate perceptions with respect to development team satisfaction and software quality in the UAM 0405 quasi-experiment

UAM 0405	Participative safety	Support for innovation	Team vision	Task orientation
Satisfaction	0.398**	0.490**	0.246**	0.499**
Software quality	0.115*		0.134*	

* $p < 0.05$ ** $p < 0.01$

preferred by the team members. The team climate categories are specific for each factor, that is, they were established based on the difference between the measures of each climate factor taken in the pre and post phases. This results in three climate preferences-perceptions fits: better than preferred, equal to preferred and worse than preferred.

The analysis of mean differences was used to find out whether there are relationships between the defined team climate categories with respect to developed software product quality and team satisfaction. To do this, we established software product quality categories based on team scores. The score categories were: C1 up to 6.99 points, equivalent to “poor” software quality; C2 from 7.00 to 7.99 points, equivalent to “acceptable” software quality; C3 from 8.00 to 8.99 points, equivalent to “good” software quality; and C4 from 9.00 to 9.99 points, equivalent to “excellent” software quality.

The same procedure was enacted for team satisfaction. The categories were defined taking into account that there are three questions for measuring satisfaction and the response values ranged from 1 to 5 points. The score categories were: C1, equivalent to “low” satisfaction (from 3 to 9 points); C2, equivalent to “good” satisfaction (from 9.1 to 12 points); and, finally, C3, equivalent to “excellent” satisfaction (from 12.1 to 15 points).

We used contingency tables to study these relationships (Acuña et al. 2008), analyzing the software product quality or team satisfaction categories with respect to the team climate preferences-perceptions fit for each climate factor. Finally, we applied Pearson’s chi-squared test to test the relationship of the software quality and satisfaction categories with respect to climate fit (better, equal, worse).

From the analysis, we found that there is a positive direct relationship between the preferences-perception fit and software quality for the team climate factors of participative safety and team vision. In other words, teams where the team climate for participative safety and team vision was better than team member preferences produced better quality software than teams that had a worse than preferred team climate in terms of participative safety and team vision. Teams whose team climate was equal to team member preferences for participative safety and team vision produced better quality software than teams whose team climate for participative safety and team vision was worse than team member preferences at the start of the project.

Finally, satisfaction in teams whose team member perceptions of support for innovation and task orientation climate factors are better than or equal to their preferences is greater than in teams whose perception of these climate factors is worse than the preferences stated by team members.

Table 12 shows the significant values after applying Pearson’s chi-squared test to examine the relationships between software quality and team member satisfaction with respect to the team climate preferences-perceptions fit (better, equal and worse).

Additionally, we used the Tukey test to calculate the differences between the means of the team climate factors for each project phase (pre, during and post). These results are shown in Table 13, illustrating the statistically significant differences ($p \leq 0.05$) between one and the same factor in each phase represented by different letters. The letters from A to H merely

Table 12 Pearson's chi-squared matrix for the team climate preferences-perceptions fit with respect to development team satisfaction and software quality in the UAM 0405 quasi-experiment

UAM 0405	Preferences-perceptions fit							
	Participative safety		Support for innovation		Team vision		Task orientation	
	Value	<i>p</i>	Value	<i>p</i>	Value	<i>p</i>	Value	<i>p</i>
Satisfaction			9.41*	0.0500			16.37**	0.0026
Software quality	18.17**	0.0058			15.45*	0.0170		

* $p < 0.05$ ** $p < 0.01$

indicate whether there are statistically significant differences between the measures taken in the different phases (pre, during and post) for each factor.

Table 13 shows that for each team climate factor there are differences between the preferences measured in the pre phase and the perceptions measured in the during and post phases. These differences are indicated by means of different letters. Repeated letters indicate that there are no significant differences between the mean perceptions measured in the during and post phases for each factor. This means that the perception of all team climate factors can be measured either during or after project development without influencing the result, because no significant differences were found. Therefore, even if one of these measurements were omitted, we would obtain the same information, irrespective of the measurement being taken during or at the end of project development.

As of this point, the remainder of the analysis uses only the team climate preference measurements taken in the pre phase and the team climate perception measurements taken in the post phase, as during-phase measurements provide no additional information with respect to the other two.

Table 13 Mean differences between the four factors and different measurement phases

Factor	Phase	Mean	Number	Significant differences = different letter	Meaning
Task orientation	Pre	24.65	35	A	Difference between pre and during and between pre and post
	During	24.14	35	B	No difference between during and post
	Post	24.16	35	B	
Support for innovation	Pre	33.82	35	C	Difference between pre and during and between pre and post
	During	30.55	35	D	No difference between during and post
	Post	32.06	35	D	
Team vision	Pre	46.37	35	E	Difference between pre and during and between pre and post
	During	44.55	35	F	No difference between during and post
	Post	44.23	35	F	
Participative safety	Pre	47.84	35	G	Difference between pre and during and between pre and post
	During	45.73	35	H	No difference between during and post
	Post	45.09	35	H	

3 Information About the Replication

The following sections detail the design of the replication according to the guidelines proposed by (Carver 2010).

3.1 Motivation for Conducting the Replication

Combining results from several experimental studies will help to generalize the evidence gathered in the original experiment. The original study was conducted at a specific university and for a specific course and needs to be replicated to test whether the findings are supported in other contexts, for example, at another university or for another course. Repeated replication should allow gradual generalization of the results into multiple contexts and, in the long term, into evidence regarding software development generally. This paper reports only one replication but aims to show the contribution of this research with a view to motivating future repetitions. When we finally have a varied set of replications of the original experiment, we will be able to move on to other types of studies, such as the aggregation of replications in order to synthesize the results and reach more conclusive findings. The replication of empirical studies is fundamental for bolstering the confirmatory power of the results (Brooks et al. 2008). As a natural follow-on from this quasi-experimental research, we are designing and conducting a controlled experiment to evaluate the validity of some of the conclusions reached and verify the causality of the relationships.

The original experiment, described in Section 2, had two goals. The first goal was to analyze the personality, team processes and task characteristics, whereas the second was to examine team climate, related in both cases to software product quality and team member satisfaction. The results of the original experiment revealed some significant relationships for two experimental goals. This convinced us that we should define a strategy to confirm these results. To do this, we replicated the empirical design of the original experiment in order to confirm and consolidate the results of the original study and obtain robust conclusions about the identified relationships.

Like the original experimental study, the replication was divided into two parts. Again, the first addressed team personality, team characteristics and team processes, and the second examined team climate with respect to software quality and team satisfaction. The goal was to describe the elements that have to be taken into account in the team formation and maintenance process to achieve better software product quality and an adequate degree of satisfaction for development process members. Table 14 shows the characterization of the original experimental study and the replication, both described in this paper.

3.2 Level of Interaction with Original Experimenters

This section explains an important aspect with regard to the interpretation of the results of this replication, that is, the level of interaction between the replicating and original researchers. By clearly defining the relationship between the people involved in the two studies, we will be able to more rigorously discuss the possible effects and consequences of interaction on the findings of the original study that we are investigating in the replication.

There is more than one experimenter role in an experiment. For example, one person may participate in the design of an experiment, another in the running of the experiment and another in the analysis. Of course, one person can perform all the roles. It has been suggested that there are five roles that can influence the result of an experiment: designer, trainer, monitor, measurer and analyst (Gómez 2012). An experiment does not have to have all five roles. For

Table 14 Characterization of the original study and replication

	Quasi-experiment UAM 0405	Replication UPM 0506
University	Universidad Autónoma de Madrid	Universidad Politécnica de Madrid
Year	2nd	5th
Subject	Data structures and algorithms	Software engineering
Participants	105	110
No. teams	35	34
Participants/team	3	3 OR 4
Task	Design & implementation	Analysis & design
Development process type	XP process adaptation	Unified process

example, if the experimental subjects do not require training, there will be no trainer; or the measurer may, depending on the type of metric used, be unnecessary. To improve the understanding of the interaction among experimenters of the original study and the replication, we analyzed the different roles within our replication:

- The designer is responsible for deciding how treatments are allocated to groups, outlining the operation of the experiment and planning the execution. The designer devises strategies for increasing control to minimize the validity threats to the experiment. The designers of the original experiment and of the replication were the same people. Therefore, the match is considered to be 100 %.
- The trainer is responsible for transmitting treatments to the subjects during training sessions; treatments can be transmitted as part of a course, seminar or workshop. The trainers were not the same people in the original experiment and the replication. Therefore, the match is considered to be 0 %.
- The monitor looks after the subjects during the experiment session. The monitor gives instructions to the subjects about how the session is to be run, hands out materials to subjects, collects in the materials and answers questions. There were two monitors in both the original experiment and the replication, and one was the same person in both cases. Therefore, the match is considered to be 50 %.
- The measurer enacts the measurement procedure on the data gathered during the experiment. This was the same person in the both original experiment and the replication. Therefore, the match is considered to be 100 %.
- The analyst conducts the statistical analyses. It is the analyst's job to identify possible outliers, prepare descriptive analyses, do significance testing, post-hoc comparison testing and other analyses. The same person did the analyses in both the original experiment and the replication in this case. Therefore, the match is considered to be 100 %.

Table 15 summarizes the number of people playing each role and the match between the original experiment and replication. If the same people played the role, the match is considered to be 100 %.

3.3 Changes to the Original Experiment

This section describes how the original experiment was changed or evolved in the replication. The replication was initially designed to respect the conditions of the original experiment whenever possible in order to assure that the results would be comparable. We realize that any change to the design of the replication can have an unforeseeable effect on the

Table 15 People and roles in original experiment and replication

Roles	Participation	Match
Designer	2 people	100 %
Trainer	2 people	0 %
Monitor	2 people	50 %
Measurer	1 person	100 %
Analyst	1 person	100 %

results. However, the replication setting is not fully controllable and adaptable. Therefore, the researchers that designed the replication have to assume that there will be differences and look out for any impact that they may have on the results that they set out to verify.

The replication of the original experiment was run by the same people; therefore, they were fully committed to and knowledgeable about the research and tried to take into account what aspects can be improved with respect to the original experiment.

In the following, we discuss the differences between the two experiments and their possible implications:

3.3.1 Participants and Subjects

Although the participants were again students, the year, degree, university and subject were different. The participants in the replication were 5th-year Universidad Politécnica de Madrid informatics engineering students in the 2005/06 academic year. These students were taking the Software Engineering II (ISII) course, as part of which student teams have to complete a practical software development assignment.

Again, students were divided into and taught as two groups (ISII-a and ISII-b). However, the activities that had to complete as part of the practical assignment and the people that played the role of trainers were different. Both groups completed the set practical assignment following the object-oriented paradigm. Two academic staff members were responsible for following up and assessing the developed projects for both groups, subject to the same criteria established at and known to students from the start of the course. The practical assignment teams were established independently of the group to which they originally belonged.

The complexity of the practical software development project was medium. The goal of the project was for students to demonstrate their knowledge of software systems analysis and design by applying the respective techniques of the object-oriented software development paradigm for the set problem. Specifically, all teams applied the Unified Process (Jacobson et al. 1999). Although they were novices, students were experienced enough to tackle the set project. Additionally, teams were required to document the project.

The total number of participants and the number of team members was also different to the original study. There were a total of 110 participants in this empirical study, of which 83 were male (75.5 %) and 27 were female (24.5 %). The students were divided into 34 teams, of which 26 had three members and 8 had four members each. Therefore, the subject of this quasi-experiment was a three- or four-member team.

3.3.2 Instrumentation and data collection procedure

From the analysis of the mean difference between team climate factors and measurement phases conducted in the original quasi-experiment, we were able to confirm that the

perception of all the team climate factors can be measured in either quasi-experimental phase without influencing the result. This means that no significant differences were found during or after project development. Even if one of these two measurements is omitted, the information obtained is the same irrespective of the measurements being taken during or at end of project development. In response to these results, the during phase was omitted from the replication, as shown in Table 16, and the measurement instruments were distributed across the phases as in the original experiment. However, this design was validated for the team climate factors only.

Due to the omission of the during phase, the measurement of conflict and cohesion was also moved to the post phase, although the possible impact that this decision might have on the study results had not been investigated. This poses a validity threat to the study. To minimize its impact, team members were asked to document their thoughts describing team conflict and cohesion issues during project development (time between the pre and post phases). This document should assure that the results are not distorted by respondents having to rely on memory. However, we did not analyze the diaries kept by team members and cannot be sure that the questionnaires realistically recorded team conflict and cohesion episodes. Therefore, the replication really focused only on the post-episode responses.

Software quality was assessed by scoring the projects developed by teams based on the submitted documentation and team member participation. The academic staff responsible for the subject supplied the scores after experts had corrected and assessed the generated products. As the tasks were not the same as in the original experiment, the metric used for scoring purposes was different too. The metric used in the replication accounts for aspects of analysis, design and participation.

The procedure for calculating the team climate preferences-perceptions fit was the same as in the original study. The statistical data were collected and analyzed in the same manner as specified in the original experiment.

4 Comparison of Results to Original

This section reports and compares the results of the replication with the original experimental outcomes. It sets out to interpret the results with respect to the two goals pursued by both studies. Through comparison we can determine the consistency between the original study and the replication with a view to drawing more reliable conclusions than if the studies were

Table 16 Distribution of the measurement instruments of replication UPM 0506

Variables	Measurement instruments	
	Pre phase	Post phase
Personality	NEO-FFI test	
Climate preferences	Team Selection Inventory (TSI)	
Climate perceptions		Team Climate Inventory (TCI)
Interdependency		Van der Vegt et al.'s questionnaire
Autonomy		Mollemans's questionnaire
Cohesion		Gross cohesion questionnaire
Conflict		Intergroup conflict scale
Satisfaction		Gladstein's questionnaire

each analyzed separately. The comparison includes two types of information. First, we report results that are similar and therefore consistent across the two experiments. This information is useful for corroborating the original theses and findings. Second, we discuss the information concerning the results of the replication that do not match the outcomes of the original study. Through this discussion we should be able to analyze possible factors that are considered to be influencing the results. In this case, the aim is to determine whether the circumstances and aspects of the original experiment that were changed in the replication are likely to be behind the differences in the results.

The following sections report the results of the replication after applying the same statistical techniques as in the original study. Additionally, we compared these results with the outcomes of the original study for each of the research goals. First, we show the analyses and interpret the results for the first goal of the study, that is, personality, team processes and task characteristics with respect to software product quality and team member satisfaction. Then we do the same for the other goal of the empirical study: team climate preferences and perceptions with respect to software product quality and team member satisfaction.

4.1 Personality, Team Processes and Task Characteristics with Respect to Quality and Satisfaction

4.1.1 Consistent Results

Table 17 reports the results of the analysis of correlations between the variables for goal 1 in the replication. In this section, we discuss only the correlations that are consistent across both studies, showing some significant associations. First, note that there is a positive relationship of the extroversion personality factor with respect to software quality ($r=0.393$ $p=0.023$), where the result of the replication matches the outcome of the original study and builds up the credibility of this relationship. Then, the alternative hypothesis H_{11_01} (There are relationships between all team member personality factors and developed software quality) is accepted for this personality factor only. On the other hand, both studies reject hypothesis H_{12_01} (There are relationships between all team member personality factors and software development team satisfaction).

Secondly, we analyze the relationships between the social conflict, cohesion and task conflict team processes with respect to team satisfaction. The first relationship is negative

Table 17 Matrix of correlations of personality, team processes, task characteristics, software quality and team satisfaction in the experiment replication

UPM 0506	Satisfaction	Software quality
Neuroticism		
Extraversion		0.393*
Openness		
Agreeableness		
Conscientiousness		
Social conflict	−0.581**	
Task conflict	−0.391*	
Cohesion	0.617**	
Autonomy		
Interdependency	0.510**	

* $p<0.5$

** $p<0.01$

with respect to social conflict ($r=-0.581$ $p=0.000$), and the second is positive with respect to development team cohesion ($r=0.617$ $p=0.000$). Additionally, task conflict has a relationship with respect to team satisfaction ($r=-0.391$ $p=0.030$). The reported results suggest that H_{12_03} (There are relationships between team processes and software development team satisfaction) is fully supported. The alternative hypothesis H_{11_3} (There are relationships between team processes and developed software quality) does not hold, as there does not appear to be any relationship with respect to the quality of the software developed by the team. As in earlier cases, these results add credibility to and are consistent with all the findings of the original study.

With respect to task characteristics, team satisfaction can be said to correlate positively only with interdependency ($r=0.510$ $p=0.002$). This result confirms that H_{12_02} (There are relationships between task characteristics and software development team satisfaction) is partially corroborated by this experimental study. Finally, we have not found any significant relationships between task characteristics, autonomy and interdependency and software quality. Therefore, hypothesis H_{11_02} (There are relationships between task characteristics and developed software quality) is rejected.

From these results, we can say that satisfaction is related to team processes irrespective of the development type used (agile method or Unified Process). Software development activities require a close interdependency among team members, who decide which is the best way to get the job done (for example, assigning people to roles, documentation, process enactment, etc.). This is only possible if the team is generous, committed to and actively involved in and has a unanimous vision of process enactment. Software development teams must be very communicative and participative to guarantee development success. Therefore, extroverted, social and participative teams have proven to build better quality software products irrespective of the development type used (agile method or Unified Process).

4.1.2 Differences in Results

Table 18 reports correlations between variables for goal 1 that were significant in the original study but did not show up in the analysis of the replication results. The most significant relationships (confidence level of 99 %) are shaded darker in Table 18 and the less significant relationships (confidence level of 95 %) are shaded lighter. The relationships are marked with + or – to indicate whether the relationship between the variables is positive or negative.

Table 18 Relationships between personality, team processes, task characteristics, software quality and team satisfaction that were found in the original study and not in the replication

UAM 0405	SATISFACTION	SOFTWARE QUALITY
Neuroticism		
Extraversion		
Openness		
Agreeableness	+	
Conscientiousness	+	
Social Conflict		
Task Conflict		
Cohesion		
Autonomy	+	
Interdependency		

For goal 1, we cannot say that team satisfaction increases in proportion to team member friendliness and attentiveness (agreeableness) and the thoroughness with which they do their job (conscientiousness), as was the case in the original experiment. Again the possible reason is the lower levels of extroversion of the participants in the replication. This reduces interaction within the team, and team members do not feel the need to build team bonds. Team members in the replication of the experiment are quite a lot more conscientious; they do not feel the need to be satisfied with their job performance or to be with their teammates, that is, it is reasonable to assume that these responsible people instinctively pull their weight and do a good job.

The difference in the relationships with respect to personality factors in the replication appears to depend exclusively on the participants. Results would be less likely to fluctuate if we analyzed a larger sample with personality factors closer to the standard mean of the test. Additionally, a larger sample would reduce the standard deviation, that is, would reduce the personality dispersion across team members.

Finally, autonomy was also found to have a positive relationship to team satisfaction in the original experiment, that is, the possibility of deciding on how to organize the team and perform tasks is positively related to the level of team member satisfaction. It could be that the subject teaching method dictates that there is no alternative work method for teams to use to perform the task in the experiment replication. Therefore, the reason that there is no such relationship may depend on the type of development used, that is, agile method or Unified Process, which is not the same in the two experiments.

4.2 Team Climate

4.2.1 Consistent Results

Table 19 shows the matrix with statistically significant linear regressions (R^2). Table 19 illustrates the relationships of team climate perceptions with respect to developed software quality and team member satisfaction for the experiment replication. In this section, we discuss only the relationships that are consistent across the two experiments, showing some significant associations. Similarly, Table 20 reports the significant relationships between the team climate fit with respect to software quality and team satisfaction.

The data in Table 19 show that, with the exception of the team vision factor with a confidence level of 95 %, all the team climate perception factors are related to team satisfaction at a confidence level of 99 %. The reported results suggest that H_{22_05} to H_{22_08} are fully supported. On the other hand, software quality is only related to the perception of the participative safety factor with a confidence level of 95 %. The alternative hypothesis H_{21_05} is supported. But H_{21_06} to H_{21_08} do not hold, as there does not appear to be any relationship with respect to the quality of the software developed by the team. These results are consistent with the outcomes of the original study and add credibility to the idea that team climate factor

Table 19 Matrix of linear regressions of team climate perceptions with respect to development team satisfaction and software quality in the experiment replication

UPM 0506	Participative safety	Support for innovation	Team vision	Task orientation
Satisfaction	0.573**	0.210**	0.131*	0.463**
Software quality	0.187*			

* $p < 0.05$

** $p < 0.01$

perceptions after project development are related to software developer team satisfaction. Therefore, team satisfaction will be greater in teams whose members perceive a team climate where all team members are encouraged and moved to participate, there is an innovative climate, a climate that supports team work, and a climate where team members are clear about the target goals, and team members will want to stay with the team.

Table 20 shows the values of the replication after applying Pearson's chi-squared test. It illustrates the relationships that are consistent across the original study and the replication. Table 20 suggests that there is a positive relationship between preferences-perceptions fit of the participative safety team climate factor with respect to the software quality categories. The alternative hypothesis H_{21_09} is supported. But H_{21_10} to H_{21_12} do not hold, as there does not appear to be any relationship with respect to the quality of the software developed by the team. In other words, teams where the team climate was better than the team member preferences with respect to participative safety produce better quality software than teams whose team climate was worse for participative safety. Teams where the team climate matched the team member preferences in terms of participative safety produce better quality than teams where the team climate for participative safety was worse than the team member preferences stated at the outset of the development project.

Finally, Table 20 shows that team member satisfaction in teams whose members have a better or equal perception of the task orientation climate factor than their preferences is greater than for teams whose perception of this climate factor is worse than the preferences stated by the team members. The alternative hypothesis H_{21_12} is supported. But H_{21_09} to H_{21_11} do not hold, as there does not appear to be any relationship with respect to team member satisfaction.

The results reported for both perceptions and for the team climate fit are consistent in both experimental studies. The consistency again adds credibility to the theses discussed in the original study and generalizes these results irrespective of whether the enacted development process was an agile or Unified Process.

4.2.2 Differences in Results

The experiment replication reproduced all the results of the original study found after analyzing the linear regressions of team climate perceptions. The only difference is that we have identified a new relationship in the quasi-experiment replication between the task orientation perception factor and developed software quality. Table 21 shows the value of the linear regression of this relationship.

Table 20 Pearson's chi-squared matrix for the team climate preferences-perceptions fit, software quality and team satisfaction in the experiment replication

UPM 0405	Preferences-perceptions fit							
	Participative safety		Support for innovation		Team vision		Task orientation	
	Value	<i>p</i>	Value	<i>p</i>	Value	<i>p</i>	Value	<i>p</i>
Satisfaction							9.70*	0.0458
Software quality	11.01*	0.0264						

* $p < 0.05$

Task orientation signifies a climate where job performance quality is a priority, and consideration is given to the professionalism with which team members do their job and any possible improvements. Therefore, teams whose members perceive a climate where all team members are encouraged and moved to participate develop higher quality software products.

Table 22 illustrates the results that did not show up in the replication with respect to the climate preferences-perceptions fit. The relationships in Table 22 are shaded lighter as they all have a confidence level of 95 %. The relationship is marked with a + sign if the relationship among the variables is positive.

We found a relationship between the preferences-perceptions fit for the support for innovation team climate factor with respect to the categories of team work satisfaction in the original study that was not identified in the replication. With respect to the developed software quality categories, the replication did not find the relationship to the team climate fit categories that was identified in the original study.

Table 23 shows the significant values after applying Pearson's chi-squared test to examine the relationships between software quality and team member satisfaction with respect to the team climate preferences-perceptions fit (better, equal and worse). We have found that the task orientation team climate factor is related to the software quality categories. In other words, teams where the team climate was better than the team member preferences for task orientation, where job performance is an important concern, produced better quality software than teams whose team climate was worse and there was less task orientation. Additionally, teams whose team climate matched team member climate preferences for task orientation produced better quality software than teams where the team climate for task orientation was worse than the preferences that team members had the start of the project.

Some of the changes made to the original study in the replication, for example, the fact that participants in the replication are students that are closer to the end of their degree and the transition to the world of work, could account for these results. This means that the task that they are set is considered to be a practical exercise closer to a professional project, and this can stimulate and motivate them to make more effort to improve its quality in preparation for professional practice.

We have also found that there is a relationship between the preferences-perceptions fit for the participative safety and team vision climate factors with respect to team satisfaction categories in the experiment replication. In other words, cohesive teams, where there is a team climate encouraging team members to participate, were more satisfied than teams whose team climate was worse than preferences, the team was not very united and team member participative safety was worse. Teams where participative safety and team vision matched the preferences of their team members were more satisfied than teams whose participative safety and team vision was worse than team member preferences at the outset of the project.

Table 21 Relationships between team climate perceptions, software quality and team satisfaction that were found in the experiment replication and not in the original study

UPM 0506	Participative safety	Support for innovation	Team vision	Task orientation
Satisfaction				
Software quality				0.117*

* $p < 0.05$

Table 22 Relationships between team climate preferences-perceptions fit, software quality and team satisfaction that were found in the original study and not in the experiment replication

UAM 0405	PREFERENCES-PERCEPTIONS FIT			
	PARTICIPATIVE SAFETY	SUPPORT FOR INNOVATION	TEAM VISION	TASK ORIENTATION
SATISFACTION		+		
SOFTWARE QUALITY			+	

5 Drawing Conclusions Across Studies

The aim of this section is to synthesize two reported quasi-experiments. To do this, we present the most important information about the original experiment and the most representative changes made in the replication that are, a priori, likely to influence the final results. Finally, we discuss the results of analyzing both experimental studies together. The aim is to interpret both studies with respect to their two goals and determine the consistency between the original study and the replication. A joint interpretation of the results means that conclusions are better explained, discussed and grounded. By reporting the global results, we will improve the visualization of the final conclusions.

The ultimate motivation for conducting this research was to draw more reliable conclusions than an independent analysis of each study would provide and dig up new information not evidenced beforehand, as well as taking the ESE methodology a step further. Let us not forget, however, that both experiments were conducted in an academic environment. For them to be generalized to an industrial setting, special-purpose quasi-experimental studies at software developer organizations would have to be planned and designed. Another aspect worth considering is team size. Both experiments were run with small-sized teams. Therefore, to generalize the results to any team size, further experiments with larger teams would have to be run. Consequently, the results are not conclusive, although they can be considered significant and, in some respects, representative of the team climate for the field of software engineering. On the one hand, these conclusions provide a better picture of which factors affect quality and team satisfaction. We can use these findings to design controlled

Table 23 Pearson's chi-squared matrix between the team climate preferences-perceptions fit, software quality and team satisfaction that were found in the experiment replication and not in the original study

UPM 0405	Preferences-perceptions fit							
	Participative safety		Support for innovation		Team vision		Task orientation	
	Value	<i>p</i>	Value	<i>p</i>	Value	<i>p</i>	Value	<i>p</i>
Satisfaction	21.75**	0.0002			13.27**	0.0100		
Software quality							9.58*	0.0481

* $p < 0.05$

** $p < 0.01$

experiments to evaluate the causality of the relationships. The natural evolution of this quasi-experimental research will be to design and run a controlled experiment to evaluate the validity of the reported findings. On the other hand, we can prepare guidelines to help to improve the formation of software development teams that produce better quality software products and whose members are more satisfied.

The following sections report each of the goals defined in the research separately. The first concerns the relationships between personality, team processes and task characteristics with respect to the quality of the software developed by the team and team member satisfaction. The second deals with the relationships between team climate and also analyses the relationships of team climate fit with respect to the quality of the software developed and team member satisfaction. Finally, we list a set of guidelines that we propose for software development team managers.

5.1 Personality, Team Processes and Task Characteristics with Respect to Quality and Satisfaction

Table 24 reports all the relationships and their significance levels for the original study and the replication. Table 24 is a matrix where the most significant relationships are shaded darker (confidence level of 99 %) than the less significant relationships (confidence level of 95 %) and the + and – signs in the cells indicate whether the relationship between the variables is positive or negative, respectively.

The results of the two experiments conducted show that software quality is not related to most of team member personality factors. Noteworthy is the positive relationship between software quality and extroversion in the original study. Note also that this relationship appears again in the replication. Hence, extroversion is likely to have an influence on the quality of the developed product, and therefore it is interesting to highlight its importance for software development team formation irrespective of the development process, agile method or Unified Process, enacted by teams.

Note that extroversion includes aspects related to cordiality, sociability, optimism, assertiveness, etc. This means that extroverts like to deal with people and find it easy to interact, interrelate and communicate with others. Therefore, it is not surprising that extroversion is related to the quality of products developed by small teams using agile or Unified Process

Table 24 Significant relationships between personality, team processes, task characteristics, software quality and team satisfaction in the original study and the replication

	SATISFACTION		SOFTWARE QUALITY	
	UAM 0405	UPM 0506	UAM 0405	UPM 0506
Neuroticism				
Extraversion			+	+
Openness				
Agreeableness	+			
Conscientiousness	+			
Social Conflict	-	-		
Task Conflict	-	-		
Cohesion	+	+		
Autonomy	+			
Interdependency	+	+		

development methods in order to produce software in a short period of time. Barrick and Mount (1991) reported the same result, where extroversion correlated positively with individual performance when the tasks involved social interaction (managers and sellers). But extroversion should be no more than average to prevent team members from getting distracted. Barry and Stewart (1997) also found that an average proportion of extroverted team members was related to successful task and team performance.

We have obtained more significant relationships with respect to team satisfaction in the two experiments conducted. Thus, satisfaction exhibits significant relationships with interdependency. This relationship should be interpreted from the viewpoint of the level of dependency among the different activities or tasks that are performed in a software project, which will again be related to the dependency between team members. Therefore, any level of dependency will have a bearing on team member satisfaction no matter what development process teams enact.

Finally, the social conflict, task conflict and cohesion team processes exhibit relationships with team satisfaction. These relationships appear in both experiments with different levels of significance. The members of a cohesive team are likely to mix well and be committed to the goals undertaken by the team. In cohesive teams, there is a relationship of trust enabling members to share out tasks with a satisfactory division of work and responsibilities within the team. On these grounds, team members are satisfied and will be happy about working together on future projects. On the other hand, the relationship with social conflict will have the opposite effects to cohesion. If any level of conflict there is among members increases, the team is unlikely to work and will break up. This does not apply to task conflict, which can be beneficial for the team provided it does not exceed levels that stop work from progressing. This means that discussing how to perform tasks and what alternative or solution looks to be the best can be positive. But, if the discussions are so drawn out as to prevent the work from being completed by the scheduled deadlines, team members will no longer consider this type of conflict to be positive, leading to social conflict and causing team dissatisfaction. This is all valid for both development types, using an agile method or the Unified Process.

Therefore, the relationships that are corroborated after conducting the replication of the original quasi-experiment for the first research goal are:

- Team member extroversion is positively and directly related to the quality of the software produced by the team.
- Team processes, task conflict, social conflict and cohesion are significantly related to team member satisfaction.
- Interdependency is positively and directly related to team member satisfaction.

5.2 Team Climate

Tables 25 and 26 summarize the results of the two experiments for team climate perceptions and the team climate preferences-perceptions fit with respect to development team satisfaction and software quality. Both tables indicate the strength and sign of these relationships. Thus, the more significant relationships are represented in darker grey (confidence level of 99 %) than the less significant relationships (confidence level of 95 %). Also the positive relationships are marked with a + sign, whereas the negative relationships bear a – sign. The results of both experiments attest to the importance of the real climate that is developed within the team and its influence on software quality and team member satisfaction.

Table 25 Significant relationships between team climate perceptions, software quality and team satisfaction in the original study and replication

	P E R C E P T I O N S							
	PARTICIPATIVE SAFETY		SUPPORT FOR INNOVATION		TEAM VISION		TASK ORIENTATION	
	UAM 0405	UPM 0506	UAM 0405	UPM 0506	UAM 0405	UPM 0506	UAM 0405	UPM 0506
SATISFACTION	+	+	+	+	+	+	+	+
SOFTWARE QUALITY	+	+			+			+

In the first place, we discuss and synthesize the results detailed above on team climate factors, preferences and perceptions. On the one hand, we conclude from the analyses conducted of the results that software quality and team member satisfaction are not related to the preferences for team climate factors. Hence, team climate preferences are not likely to have a bearing on the team doing a better and more satisfactory job at developing the project.

On the other hand, however, the same analyses conducted in both experiments show totally different results for perceptions. Both software quality and team member satisfaction are very significantly related to perceptions of team climate factors.

The team climate perceived by team members does not appear to influence software quality as much as team member satisfaction, as shown in Table 25. Participative safety is the only team climate factor that is found to have a bearing on the quality of the software developed by teams in both the original study and the replication. This means that software quality can be improved by promoting a safe climate, where it is easy to share information, and put forward ideas and solutions, without criticism from others deterring new initiatives. Hence, trust should be inspired among team members so that they can participate and state their opinions and ideas to other team members. This reasoning can be extended to team member satisfaction, that is, when the team gives out clear signs of the climate being safe, its members will feel comfortable not only about contributing but also about risk taking, putting forward more, innovative ideas to the team.

The perceptions of the four team climate factors are related to team member satisfaction. Consequently, we consider it important to encourage these team climate factors to achieve team member satisfaction.

Finally, we discuss and summarize the results for the team climate preferences-perceptions fit (Table 26). Although the relationships of the team climate fit for different team climate factors with respect to software quality are not consistent, it appears that the

Table 26 Significant relationships between team climate preferences-perceptions fit, software quality and team satisfaction in the original study and the replication

	P R E F E R E N C E S - P E R C E P T I O N S F I T							
	PARTICIPATIVE SAFETY		SUPPORT FOR INNOVATION		TEAM VISION		TASK ORIENTATION	
	UAM 0405	UPM 0506	UAM 0405	UPM 0506	UAM 0405	UPM 0506	UAM 0405	UPM 0506
SATISFACTION		+	+			+	+	+
SOFTWARE QUALITY	+	+			+			+

expectations stated in the team member preferences for participative safety should be met or surpassed in order to achieve quality software products.

Teams in which the real climate is more or less equal to team member climate preferences appear to develop a product of equal or better quality. The same applies if the real team climate is better than preferences. Therefore, software product quality can be improved if a team climate is promoted that is equal to or better than team member preferences regarding issues such as information sharing, proposing solutions and ideas without detriment from other team members (participative safety).

Team member satisfaction is related to the preferences-perceptions fit for task orientation in both experiments. The other factors behave differently in the two studies. If the team member expectations stated in the team climate factor preferences for task orientation are satisfied or surpassed, team members will be more satisfied with the work method, generated climate and goal achievement.

Consequently, all this suggests that team satisfaction is higher in teams in which the real climate is more or less equal to team member climate preferences. The same applies if the real team climate is better than preferences. Therefore, team member satisfaction is likely to improve if the real team climate is equal to or better than team member climate preferences with respect to issues like professionalism and job performance improvement (task orientation).

Therefore, the relationships that appear to be corroborated by the replication of the original quasi-experiment for the second research goal are:

1. With respect to a team climate perceptions:

- The team climate perception for the participative safety factor is positively and directly related to the quality of the software produced by the team.
- The team climate perception for all factors is related to team member satisfaction.

2. With respect to the team climate preferences-perceptions fit:

- The team climate preferences-perceptions fit for participative safety factor is related to the quality of the software produced.
- The team climate preferences-perceptions fit for task orientation factor is related to software development team satisfaction.

5.3 Recommendations on Team Formation

The aim of this research is to set out the theoretical knowledge gained from the running the original empirical study and replication described above into recommendations applicable to team formation for software development in an academic setting. Both the possibility of conducting replications that strengthen the findings in the business setting and designing controlled experiments to check causal relationships would make it possible to generalize these recommendations to the software industry. These results should serve to motivate other researchers (and ourselves) to continue researching the relationships among personality, team processes, task characteristics, software development team climate with respect to software quality and team satisfaction. In order to understand the causal relationships among these aspects, it is necessary to run families of controlled experiments. Even if team managers were to consider the relationships identified in our research to form development teams, there is no guarantee that such teams would produce better quality software or that their members would be more satisfied. Even so, it will reduce the chances of producing low quality software and team member dissatisfaction.

The hypotheses tested and the conclusions drawn from the empirical studies conducted as part of this research with respect to personality, team processes and task characteristics support the proposal of empirical guidelines applicable by software project managers with respect to team formation. These guidelines are:

1. Identify the personality of potential software team members in order to determine the team extroversion level. We have found that this personality factor is related to team performance and software product quality. The NEO-FFI personality test is suitable for this purpose.
2. Encourage activities that boost communication, optimism and assertiveness within the team. This will improve team extroversion and interpersonal relationships, which may improve the quality of the developed software.
3. Project type and characteristics permitting, set interdependent tasks and implement work methods that are conducive to interdependency. This type of task appears to positively relate to team member satisfaction. Interdependency indicates that tasks are related and cannot be developed separately and independently. This also implies a relationship among people performing this task or activity.
4. Determine team process (social and task conflict and cohesion), task interdependency and team member satisfaction levels. The intergroup conflict questionnaire, gross cohesion questionnaire, Van der Vegt et al.'s questionnaire on task interdependency and finally Gladstein's questionnaire on satisfaction are useful for this purpose. It is important to monitor these factors in order to check that task conflict and social conflict levels do not reduce development team member satisfaction.

The results of the empirical studies carried out as part of this research into team climate with respect to software quality and team member satisfaction confirm that:

- The preferences-perceptions fit for participative safety is related to the quality of the software produced by the team.
- The preferences-perceptions fit for task orientation is related to team member satisfaction.

The empirical guidelines that we propose for application by software project managers with respect to team formation according to an adaptation of the agile software development method and the Unified Process are:

1. Identify software developer team climate preferences. The TSI test is designed and useful for this purpose. Although the particular preferences of developers with respect to team climate do not appear to have major implications for software product quality or team member satisfaction, they have to be known to analyze how developer expectations evolve with respect to the generated team climate.
2. Project type and characteristics permitting, collect data during project development to measure and find out what the team climate is really like (team climate perception). The TCI test is designed and can be used for this purpose. Finding out how all team climate factors are really perceived within the team is important and has implications for team member satisfaction. Finding out how the team climate participative safety factor is really perceived within the team is important and has implications for software quality.
3. Take measures (if necessary) to improve real team climate (team climate perception within the team) for the participative safety factor. There is some indication that this reduces the possibility of producing low quality software.
4. Check whether the team climate (climate perception within the team) satisfies (or is better than) team member preferences for participative safety.

If the team climate preferences-perception fit is not satisfactory, then take steps to check that the real team climate (team climate perception) for the participative safety factor is not worse than team member preferences for this factor.

Anything that is conducive to team climate preferences for participative safety being met or improved reduces the prospects of producing low quality software.

5. Check whether team climate (team member climate perception) for the task orientation factor is equal to (or even better than) team member team climate preferences.

If the team climate preferences-perception fit is not satisfactory, then take steps to check that the real team climate (team climate perception) for the task orientation factor is not worse than team member preferences for this factor.

Anything that is conducive to team climate preferences for the task orientation factor being met or improved reduces the prospects of team members being dissatisfied and wanting to leave the team.

6 Conclusions

This paper reports the replication of an empirical study applying the guidelines proposed by Carver (2010) with a view to their improvement and standardization. The following are our suggestions for improving the reporting of information concerning replications of experimental studies. We have included two aspects in this paper that we consider to be necessary for reporting any research: an introduction and conclusions. On this ground, we have added the introduction as a section describing the context of the research for readers. In this particular case, it also explains the two aims of the paper: implement the guidelines for reporting replications and analyze the results of the replication and draw cross-study conclusions for both experiments. The conclusions have a similar justification. They are useful for outlining suggestions after applying the proposed guidelines, as well as the conclusions of the experimental studies.

Additionally, as a result of this research, we have been able to list some recommendations for improving team formation, the climate of the team developing software products and software engineer satisfaction irrespective of the enacted software development process. The study is based on the two goals defined in the design of the original empirical study. First, we analyze and draw conclusions about the relationships between developer team personality factors with respect to software quality and team member satisfaction with the aim of supporting the results of the original study. Second, we study and draw conclusions about the relationships between the team member climate preferences-perceptions fit for the purpose of checking the results of the original study. Additionally, the original study and replication each used a different type of development process. Therefore the results can be analyzed from the viewpoint of the development process enacted by teams.

The results of the replication corroborate some of the findings of the original study. First, the results revealed that there is a significant correlation between the extroversion personality factor and software quality, but none with team satisfaction. Additionally, we found that team processes, task conflict, social conflict and cohesion and task interdependency are related to team member satisfaction. With respect to team climate, we found, however, that the team perception of high participative safety correlates with better software. On the other hand, we found that the four team climate factors are significantly related to team satisfaction. Additionally, the results show that there is a direct and positive relationship between teams with a better than or equal to preferred climate for the participative safety factor and software quality, whereas team satisfaction is greater for teams with equal to and better than

preferred climate for the task orientation climate factor than teams with worse than preferred climate.

These findings are useful for proposing guidelines for team managers on how to form more effective teams. These guidelines set out how interpersonal relationships within the team should be taken into account to increase the prospects of producing quality software. They also indicate that software quality and team member satisfaction is likely to be higher if team members sense that the team climate is to their liking. The enacted software process (agile method or Unified Process) has no influence.

Software engineering technology transfer is far from straightforward. Software organizations should carefully manage the adoption of these guidelines, which should be properly explained to stakeholders. The adoption of the changes to the software process and human resources management will require an extra effort on the part of software organizations.

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Marta N. Gómez is an assistant professor at the Universidad CEU San Pablo's Computer Science Department. She received her PhD in computer science from the Universidad Autónoma de Madrid. Her research interests include person-team fit and empirical software engineering. She's a member of the IEEE Computer Society. Contact her at Departamento de Tecnologías de la Información, Escuela Politécnica Superior, Universidad CEU San Pablo, Campus de Monteprincipe, 28668 Boadilla del Monte, Madrid, Spain; mgomez.eps@ceu.es.



Silvia T. Acuña is an associate professor in the Universidad Autónoma de Madrid's Computer Science Department. She received her PhD in computer science from the Universidad Politécnica de Madrid. Her research interests include software process modeling, person-team fit, empirical software engineering and usability engineering. She's a member of the IEEE Computer Society. Contact her at Departamento de Ingeniería Informática, Escuela Politécnica Superior, Universidad Autónoma de Madrid, Calle Francisco Tomás y Valiente 11, 28049 Madrid, Spain; silvia.acunna@uam.es.